

Model 60L

- 1) Select the Bookmark option
- 2) Click on appropriate Service Notice

Service Notice Numbering and Tracking System

The following layout is using a single part number for reference

65L614 Rubber Plug

8 characters used as a "File Name" on the computer system.

060 14 001

Subdivision related to the part "14" which will continually increase as the number of Lubrication Notices are created.

Type of part - Main Division, such as "14" for Lubrication or "44" for Shafts or "00" for an assembled component.

Designates Seamer Model Number

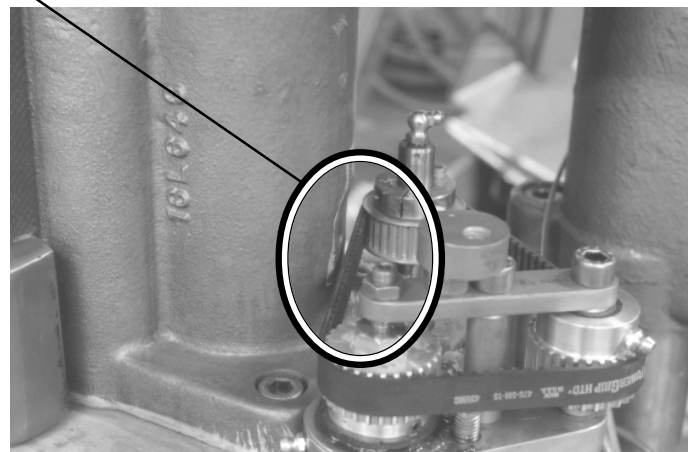
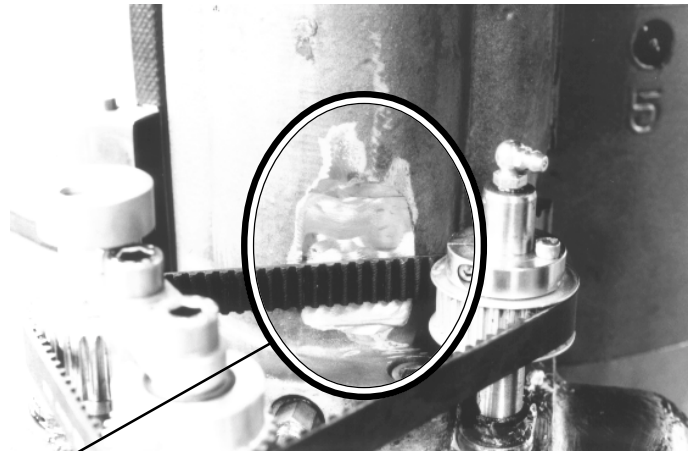
Revisions to existing Service Notices will be as follows: The "File Name" will be duplicated on the Service Notice with the latest revision level designated by a letter.

Example: 06014001"A". Revision levels on Service Notices are rare but can happen. The Service Notice will provide remaining needed information.

Model 60L Single Separator 401 to 404 Diameter Ends

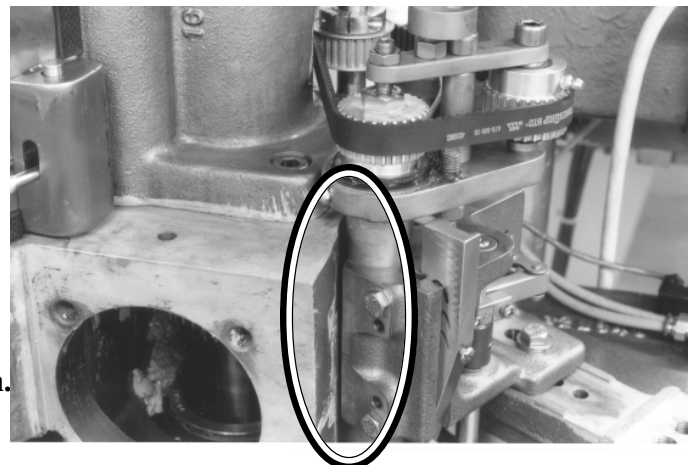
**Modifications required to
adapt the Single Separator
in the field using these diameters.**

1. The Drive Case Support Bracket casting may have to be ground out for Drive Belt clearance. This area is approximately 1/8 inch deep 2 inches wide. These dimensions may vary from one casting to the next.



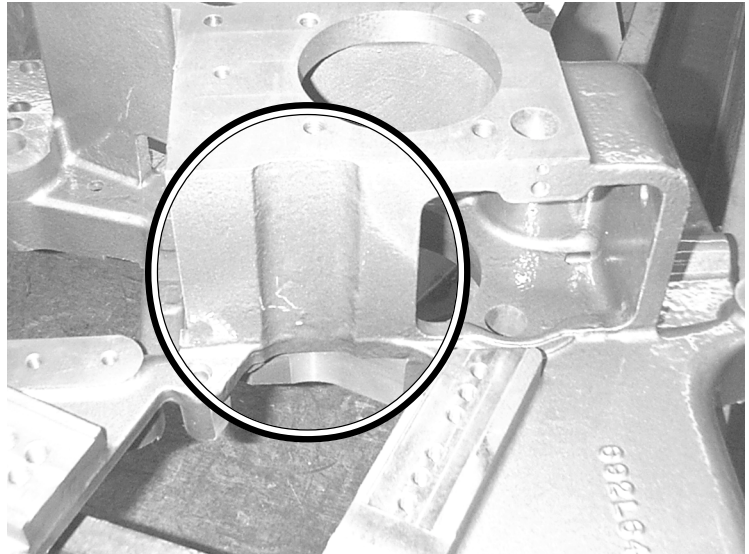
2. The Cap Feed Bracket casting will have to be ground out to allow the Separator Body to position back far enough for setting up End clearances required during assembly of the unit.
This clearance should be ground out following a similar radius as the Separator Body.

See the next page dimensions for this area.

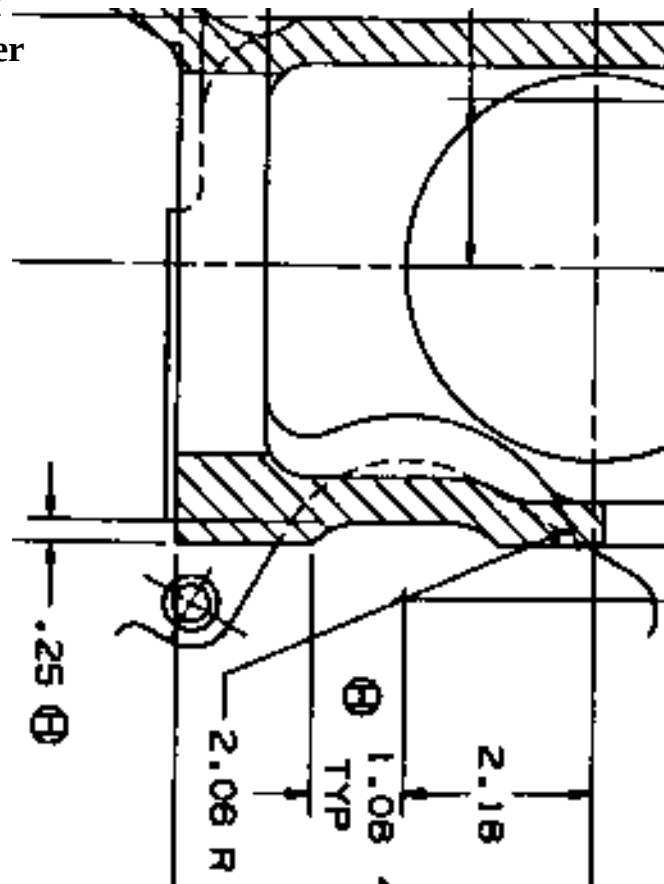


Model 60L Single Separator 401 to 404 Diameter Ends

Modifications required to adapt the Single Separator in the field using these diameters.



These areas shown will require grinding or machining prior to installing the Separator Bracket Assembly for 401 to 404 diameter covers.



**MODEL 60L
LIFTING
AND
MOVING
SUGGESTIONS**

Model 60L Seamer Lifting and Moving

4/22/98-06000002

LIFTING AND MOVING THE MACHINE

Receiving

Insure that the Truckers documents are in order and that any special shipping instructions have been followed.

Visually check each crate for obvious external damage.

When Angelus supplies transportation for domestic Seamer delivery, it is requested of Trucking firms to cover all crates with a suitable tarp before going on the road. This precaution is to protect against the outside elements during shipment.

Offloading

The Seamer crate is designed for Fork Lifts rated at a **minimum** of 7,000 lb. capacity with a minimum arm length of 6 ft.. Center load or balance markings (usually a red spray painted dot located at the bottom of the platform) are noted on all crates to even the load.

Crate Removal

Removing crates and System Enclosures from around the Seamer Enclosure must be done with care to avoid any damage to equipment inside. The largest and heaviest crates should be positioned where they can be easily accessed from all sides. Floor plan layouts provide locations for all components. Installation procedures are prepared in sequence through these instructions.

Inspection

After crates have been unpacked, the contents must be visually inspected thoroughly to detect any external damage to equipment.

1st Detect if any damage is present and start claims if necessary.

2nd Contact Angelus immediately for replacement parts.

Check inside the Electrical Panel, Remote Operator Station, Undercover Gassing Enclosure and the Lubrication Enclosure for components that may have vibrated loose during shipment.

Moving across the floor

Moving the Seamer while still on the skids it was shipped on is, in most cases, the safest way when machine dollies or a Fork lift is used.

Floor machinery dollies

Position machine dollies to balance the load at all four corners of the direct underlying weight. All machines are marked with a load balance on the crate or skid. Control horizontal movement when moving down an incline.

Fork Lift

Fork Lifts must be rated to handle a minimum 7,000 lbs. (3150 kg) on a single lift of just the machine and skid. Export crated machines can weigh up to 6,300 lbs. (2835 kg) before sides and top are removed. Center the loads.

Fork Lift Feet No. J-14198 are designed to bolt up to the bottom of each one of the Seamer Legs to make moving the machine easier and safely.

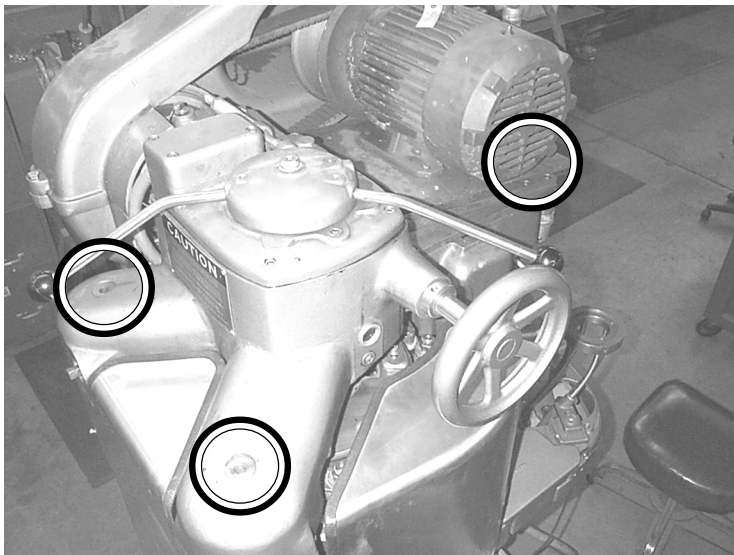
4/22/98-06000002

Fork Lift Cribbing (blocking or fixtures)

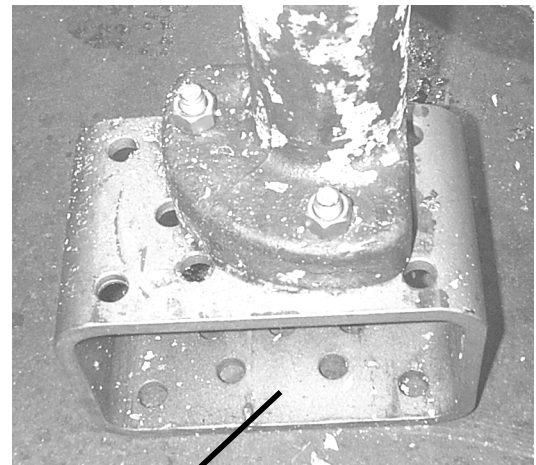
When a Forklift is used to remove the underlying skid, care must be taken to crib from forks up to the Base Housing. Lubrication Lines and the painted surface of this machine are important to not damage, bend or chip when lifting the machine. When a fork lift is used to position the machine, wood blocking is suggested between forks and the bottom of the machine to ensure it is level.

Overhead Cranes

Overhead cranes must be capable of handling specified weights. Cables or chains must be carefully located using 5/8-11 eyebolts at three separate positions in the top of the machine. The Angelus Overhead Lifting fixture J-12413 is designed to properly fit these three areas and center the load of the machine only (no can feed extension attached). The length of cables will determine how level the machine will lift. It is not suggested to lift the machine from the Base area using hooks. First of all, the machine is very top heavy and secondly although minor, the paint will chip from lifting hooks leading to severe oxidation.



5/8-11 Eyebolt locations over the Drive Case Supports and under the Motor Base. The Motor Base must be moved all the way in the expose the third area.



Fork Lift Feet #J-14198 are available to attach under each Base Leg.

WARNING:

These instructions are offered in good faith as typical reference and not as specification or requirements. The information contained herein is based on available data at the time of release. There is no warranty, expressed or implied, regarding the accuracy of this data, or the results to be obtained from the use of. Angelus assumes no responsibility for damage from the use of the information inscribed within. Moving machinery of this type should only be performed by trained and qualified personnel using appropriate equipment.



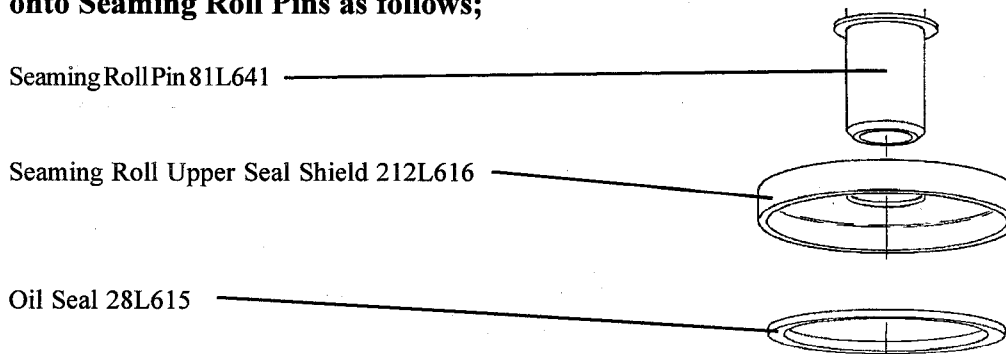
Model 60L Seaming Rolls
with
Ceramic Ball Bearing Assembly
Installation Requirements

Seaming Roll with Ceramic Ball Bearing Assembly Installation Sequence for 60L Seamers

NOTE: Seaming Rolls with Ceramic Ball Bearings are shimmed at the factory for proper running clearance before shipment. Install Seaming Roll Assemblies as packaged for shipment.

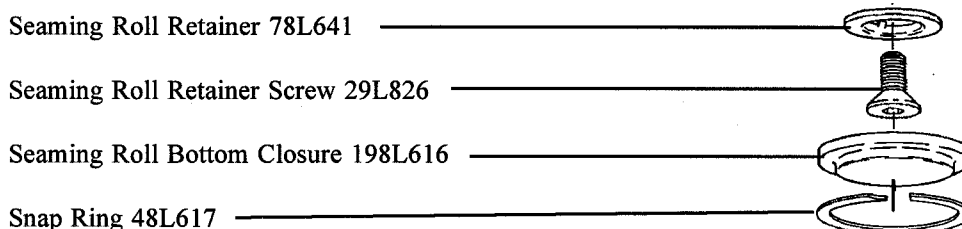
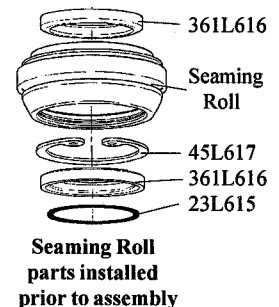
Do not mix or re-arrange assembly components.

**Install Seaming Roll components
onto Seaming Roll Pins as follows;**



**Pre-shimmed
Seaming Roll
with Ceramic
Bearings**

Ball Bearing Inner Race 360L616
1st Operation Seaming Roll 291L641 or
2nd Operation Seaming Roll 292L641
with (2) 361L616 Ball Bearing Outer Race
(1) 45L617 Snap Ring
(1) 23L615 O-Ring
Snap Ring 49L617
Shims 362L616 (as required)
Ball Bearing Inner Race 360L616



LUBRICATION REQUIREMENTS: Ceramic Ball Bearings assembled within the Seaming Roll and mounted ready for production, should only require one shot of grease from a hand grease gun for each assembly. If more than one shot of grease is installed, the ceramic ball bearing will expel excess grease out of the assembly.

Suggested lubrication intervals = every 72 hours w/N.L.G.I. Type O grease or 1 shot of grease after sanitation to remove water from the Seaming Roll.

Subject:

60L Modification to Pressurize Seaming Turret.

Purpose:

To eliminate condensation from steam inside of seaming turret by pressurizing turret with air.

Benefit:

Prolongs life of seaming turret internal parts (bearings, seaming lever shafts, seaming spindles, etc.) which in turn decreases downtime improving productivity.

Description:

Pressure Parts List (60LM128) consist of water separator (auto drain), air regulator (max. 3 PSI), pressure gage (0-5 PSI), mounting brackets, air pressure line fitting kit, seaming turret fitting kit, air supply regulator & water separator fitting kit.

**Note:**

- For Seamers without hand clutch **735L643** mounting bracket is used and installed to drive case cover (**Driven Lifters**).
- For Seamers with hand clutch **736L643** mounting bracket is used and installed to drive case (**Standard Lifters**).

Installation Procedure:**(Standard Lifters)**

- 1) Remove seaming levers.
- 2) Remove 9L654 drive case cover.
- 3) Remove 27L623 drive gear clamp ring.
- 4) Remove 18L617 intermediate drive pinion bearing retainer.
- 5) Remove 8L644 vertical drive shaft downward.
- 6) Remove 15L641 seaming cam hub and 1L620 seaming cam (raise seamer to 25" between bottom of drive case support to base).
- 7) Modify 15L641 seaming cam hub per Figure 1.
- 8) Modify 7L643 drive case per Figure 2.
- 9) Reassemble in reverse order.

(Driven Lifters)

- 1) Remove seaming levers.
- 2) Remove 9L654 drive case cover.
- 3) Remove 27L623 drive gear clamp ring.
- 4) Remove 148L823 intermediate drive pinion (allen screw).
- 5) Remove 474L644 vertical drive shaft upward.
- 6) Remove 15L641 seaming cam hub and 1L620 seaming cam (raise seamer to 25" between bottom of drive case support to base).
- 7) Modify 15L641 seaming cam hub per Figure 1.
- 9) Reassemble in reverse order.

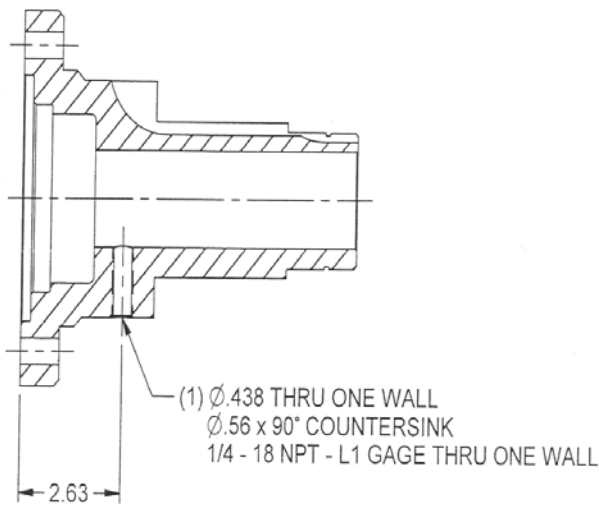


Fig. 1 Modification Required for 15L641 Seaming Cam Hub

Ref. Data Sheet G60.255

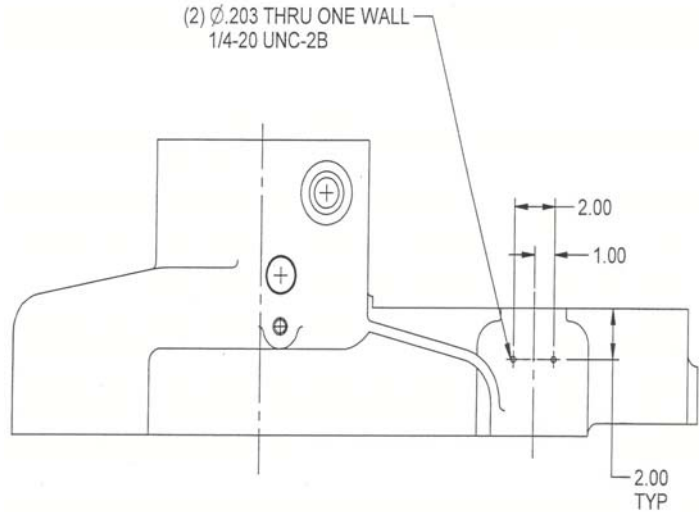
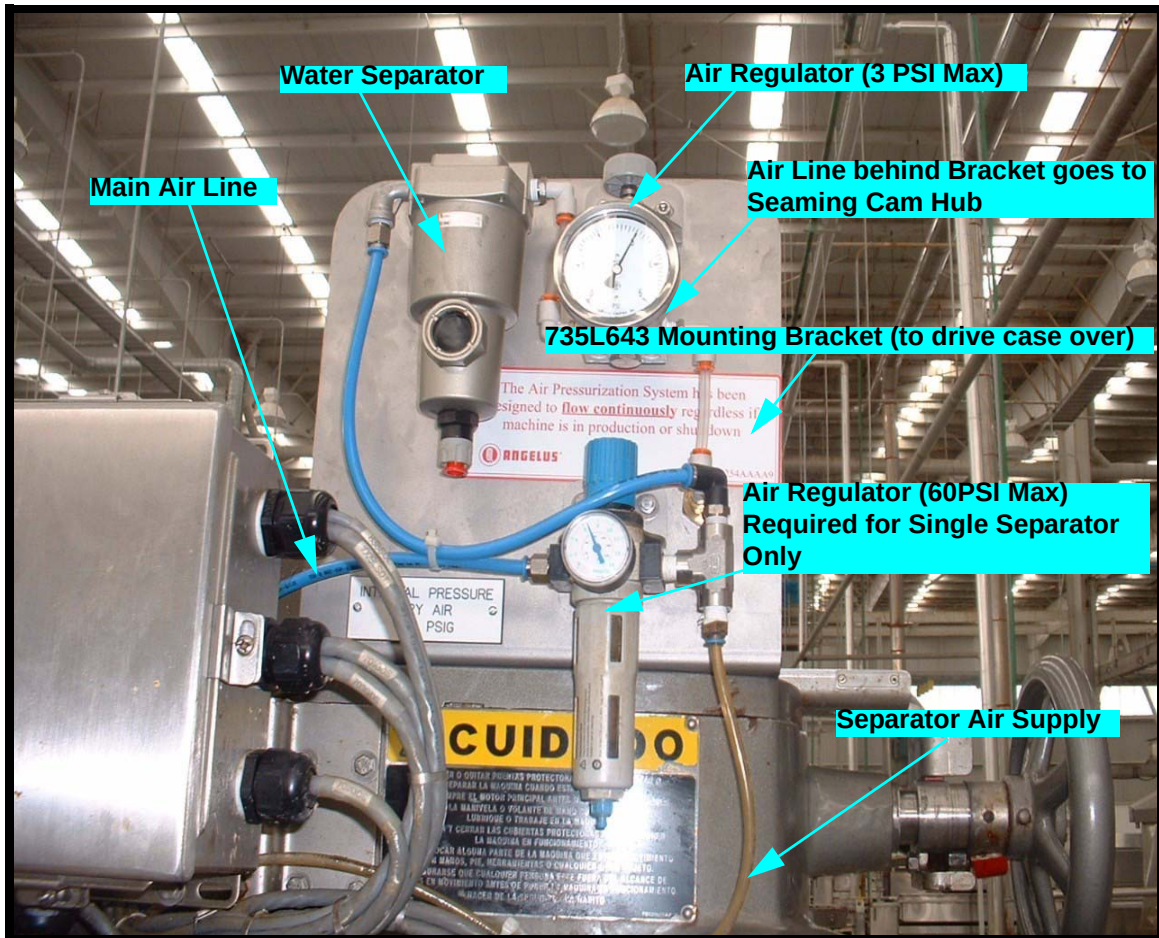


Fig. 2 Modification Required for 7L643 Drive Case only if 736L643 Mounting Bracket is used

Ref. Data Sheet G60.256





Model 61H, 62H and 60L Seamers

Subject: 65L614 Rubber Plug

Used on Seaming Roll Levers
24H641 = 62H Seamers
5H641, 6H641 and 9H641
Seaming Roll Shaft and Lever
Combination = 61H Seamers
83L641 = 60L Seamers

Description:

The Rubber Plug is used to keep grease from flowing around the Seaming Roll Pin groove and escaping through the saw cut of the Roll Lever. The Rubber Plug should be a snug fit within the 1/4-28 tapped hole and requires a certain assembly method.

How often does this need replacement?

Replacement of the Rubber Plug should be done at every overhaul of the Seamer or any time there is a rebuild of the Seaming Roll Levers themselves. Replacement at any other time would take place if there is a noticeable leakage at the Seaming Roll lubrication interval.

Installation Procedure:

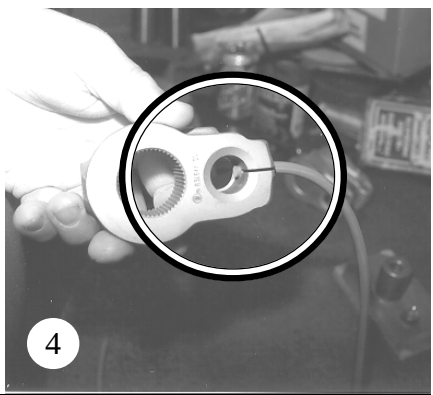
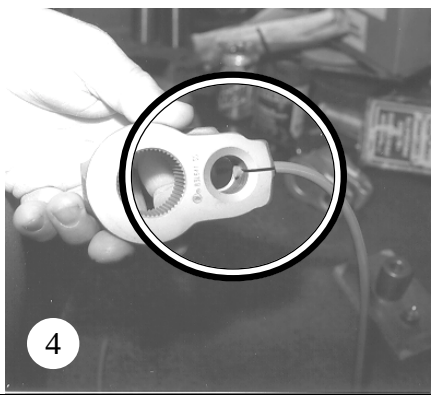
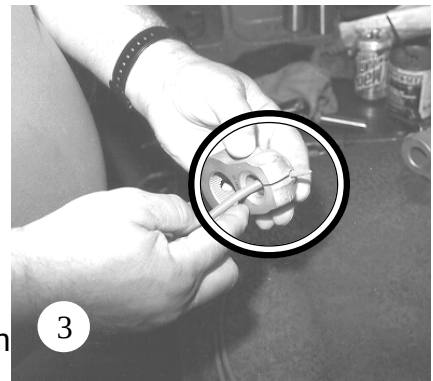
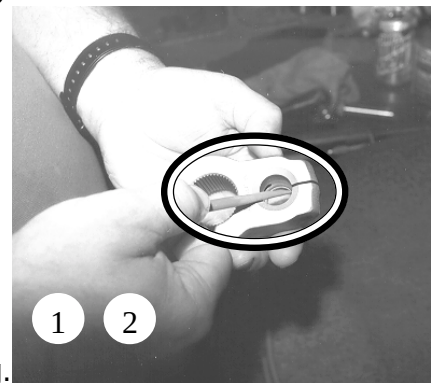
1st. Cut off Rubber Plug stock no longer than 12 inches long.

2nd. Use a razor blade, sharp knife or belt sander to taper the end of the Plug from pointed back one inch to the original diameter.

3rd. Start the Rubber Plug tapered end through the inside Roll Pin bore.

4th. Carefully pull the Plug material through the 1/4-28 tapped hole and leave approximately 1/8 inch of material on the inside of the bore. Slide the brass plug provided inside the bore under the rubber plug.

5th. Use Angelus Trim Fixture 375L616 designed to trim the Rubber Plug concentric to the bore. The brass plug saves cutter edge and makes a clean cut.



Subject:

Seven-Point Lubrication Modification for Angelus 60L machines equipped with Single Point Lube-Driven Lower Chucks.

Purpose: Angelus is announcing a simple modification that will reduce the total amount of grease required for each grease cycle for the lower chucks and center bearing.

Benefit: Reduce the amount of lubricant usage for each lubrication cycle and cut down maintenance time.

Description: Modify Angelus 60L lower seaming turret equipped with single point lube driven lower chucks to a seven-point lube system. The seven lubrication points include one point for each chuck and the additional point for the seaming turret lower bearing. This conversion does not require the removal or disassembly of the seaming turret. Estimated time for change over is two to four man-hours.

Availability:

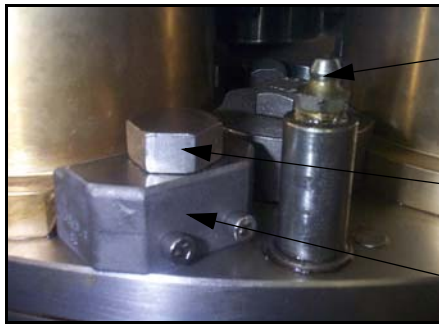
Presently, conversion parts per parts list 60LM124 are available. In the near future a revised parts list 60LM124.01 will be available. The only difference is the 140L614 lube line bolt, 3C015 o-ring, 85S815 o-ring are not needed because the 134L614 divider block is being replaced with 141L614 divider block (investment casting). The seven point concept remains the same.

Parts List 60LM124	Parts List 60LM124.01
(1) #1961S SS Alemite Fitting (6) 90° #1961S SS Alemite Fitting (12) 8-32 x 1-1/2" Soc Head Screws (1) 1/8-27 SS Hex Soc Pipe Plug (1) M12-1.75 x 70mm long Soc Head Screw (1) M-12 SS Dome Nut (6) O-Ring..... 3C015 (6) Divider Block..... 134L614 (6) Gasket Divider Block..... 135L614 (6) Lube Line Bolt..... 140L614 (6) O-Ring..... 85S815	(1) #1961S SS Alemite Fitting (6) 90° #1923S SS Alemite Fitting (12) 8-32 x 1-1/2" Soc Head Screws (1) 1/8-27 SS Hex Soc Pipe Plug (1) M12-1.75 x 70mm long Soc Head Screw (1) M-12 SS Dome Nut (6) Divider Block 141L614 (6) Gasket Divider Block..... 135L614

Installation Procedure:

(1) Remove from the top surface of the lower turret top:

- (1) Single point Alemite fitting with coupling and nipple
- (6) 133L614 lube line bolts
- (6) Original 134L614 divider blocks



Alemite Fitting

133L614 Bolt

134L614 Divider Block

(2) Remove lines and fittings underneath the lower turret bottom, including the feed line for 8H616 turret bearing. Remove pipe plug from turret periphery.



Remove:

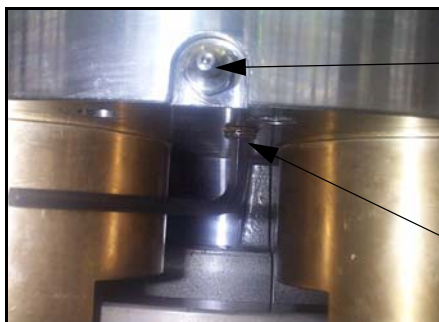
Pipe Plug

Fittings

Lines

Bearing Feed Line

(3) Under the lower turret bottom where the bearing feed line was, install (1) 1/8-27 SS hex soc pipe plug (permatex applied). On the turret periphery, install (1) #1961S SS Alemite fitting (permatex applied). This becomes the **seventh lubrication point** for the turret bearing.



#1961S Alemite Fitting

1/8-27 Pipe Plug

Note: If Topper is used; Change #1961 Fitting to 1/8-27 Pipe Plug

Change 1/8-27 Pipe Plug to #1923S 90° Alemite

(4) The pre-assembled (6) 134L614 modified divider blocks (Rev. C) assemblies consist of:

- (1ea) 3C015 O-Ring
- (1ea) 85S815 O-Ring
- (1ea) 90° #1923S SS Alemite Fitting
- (1ea) 140L614 Lube Line Bolt (never-seez applied)

Apply silicone to the bottom surface of the 134L614 modified divider blocks. Install (1 ea.) 135L614 gasket between the lower chuck slide and divider blocks and secure blocks with (2ea) 8-32 x 1-1/2" soc head screws (never-seez applied). To seal the single point hole in the turret, install (1) M-12 SS dome nut on the top side of turret and (1) M12-1.75 x 70mm long socket head screw from bottom of turret (blue Loctite applied).



90° Alemite Fitting

140L614 Bolt

134L614 Divider Block

M12 Dome Nut

Lubricate in sequence, the chucks positioned at the **high level** of the lower chuck cam. Use as needed the hand wheel to access all (6) 90° Alemite fittings plus (1) lower turret bearing Alemite fitting.

Apply grease every 8 hours until grease is visible at the indicator point on each cam roll pin.

Apply 5 shots of grease every 8 hours to the lower turret bearing.

Warning: Grease will not be delivered to all required lubricating points if chucks are in the transition areas of the lower chuck cam during lubrication attempts.

Ref. Engr. Bulletin #86

Subject:

Model 60L modification required on seaming turret bottom to install stainless steel wear washers above seaming levers.

Purpose:

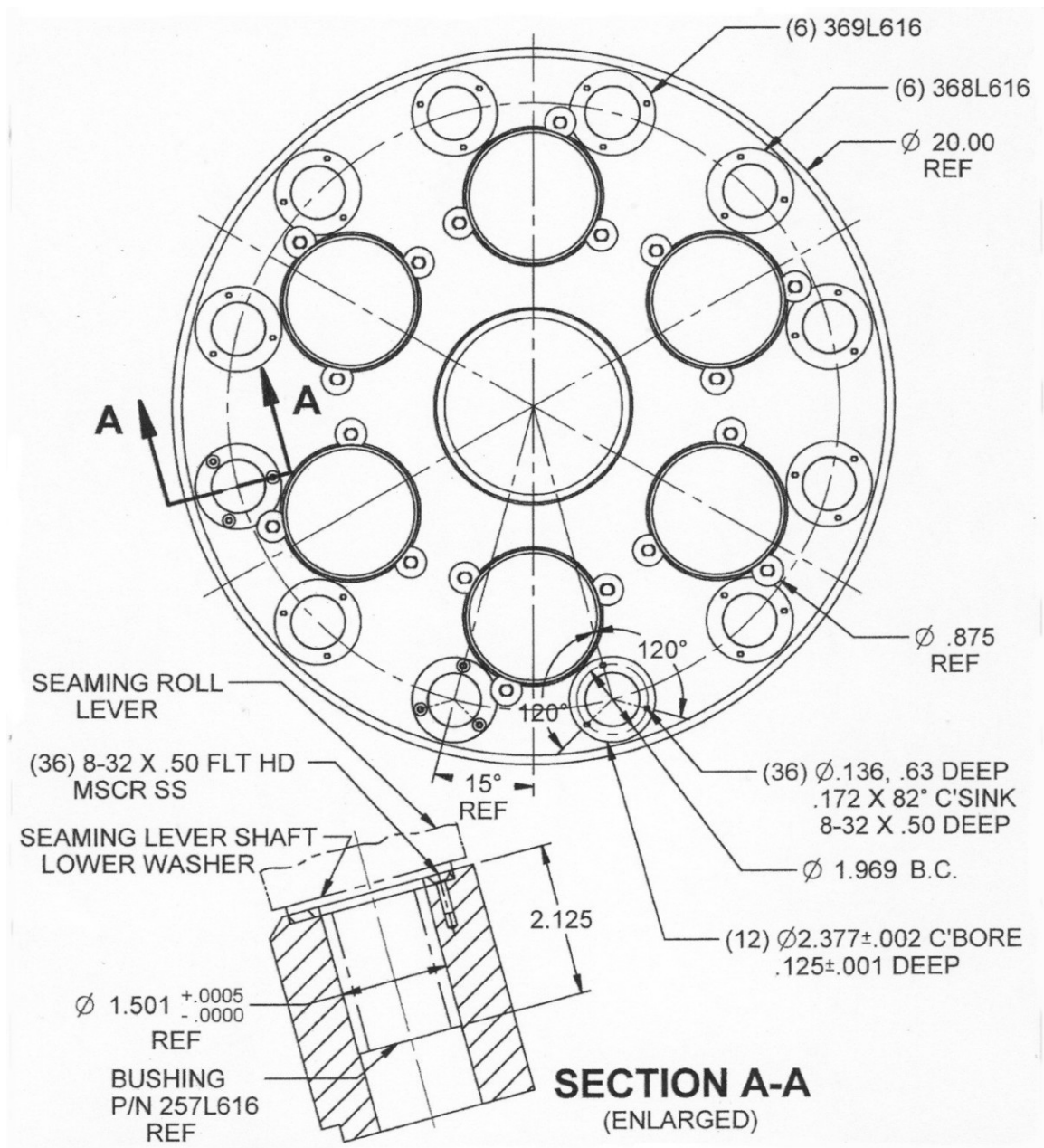
To repair the upper turret from wear of seaming lever thrust washer.

Benefit:

This modification will provide a protective barrier to the upper turret from wear and also maintain seaming roll height.

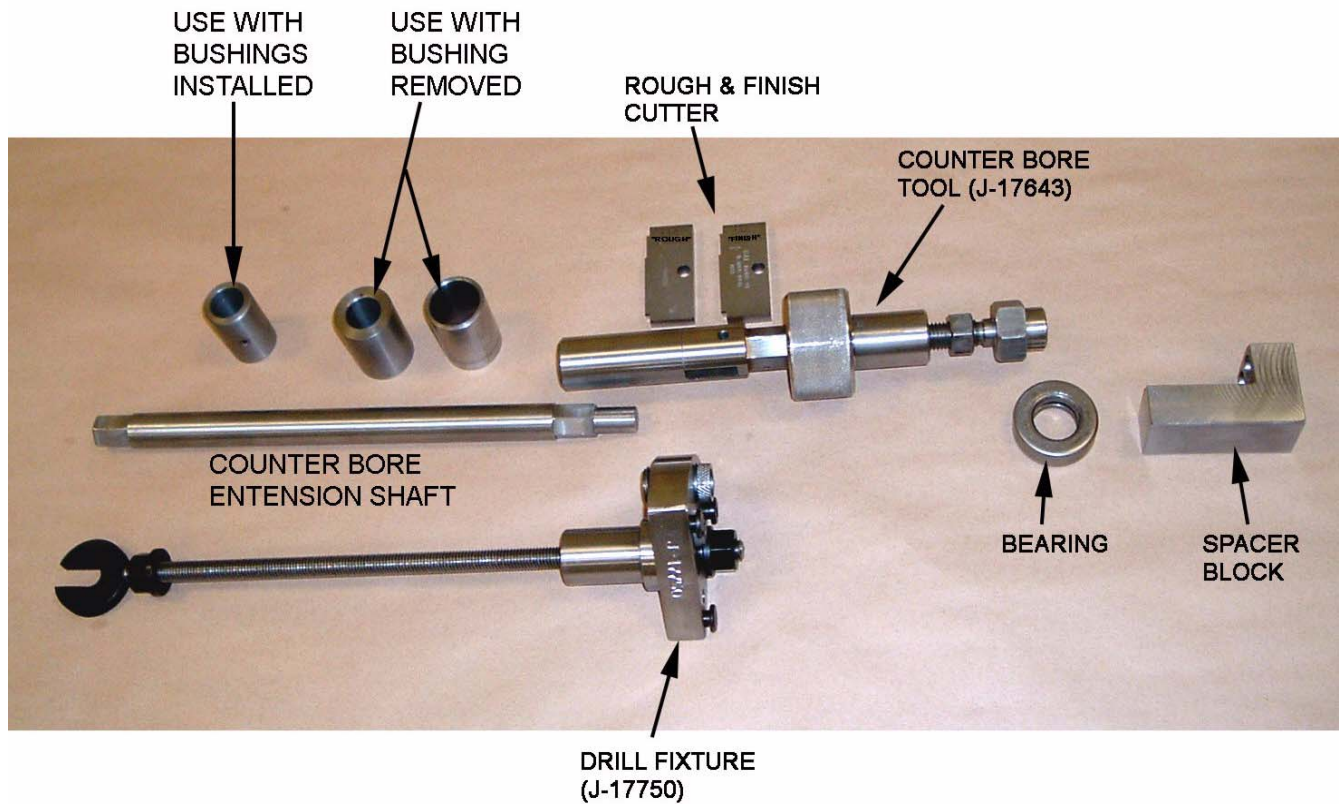
Description:

Ref. Engr. Data Sheet G60.218 (see Installation Procedure)

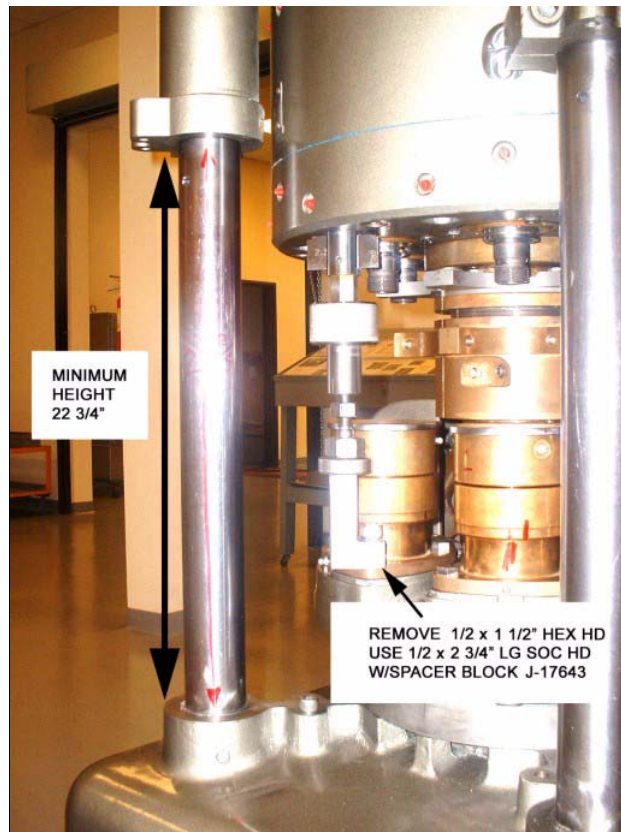


Installation Procedure:

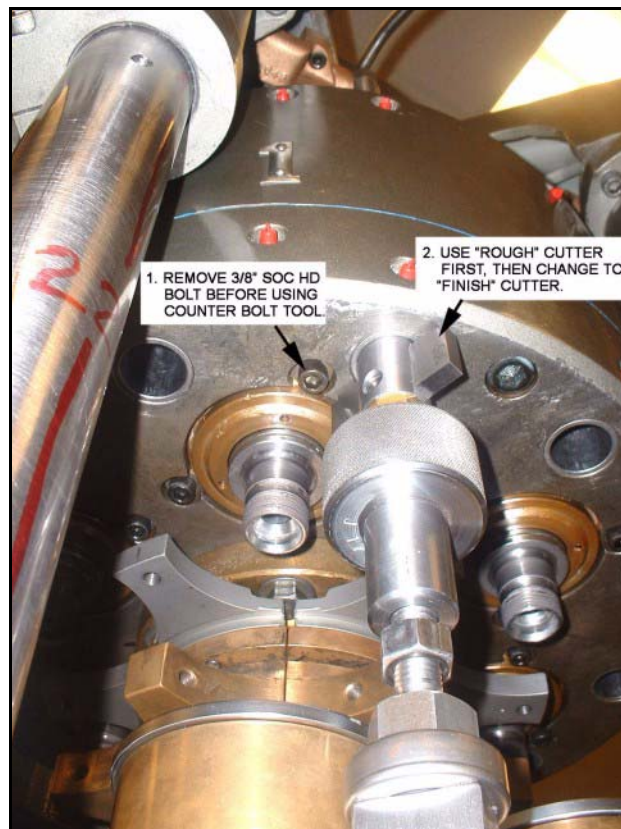
(1)



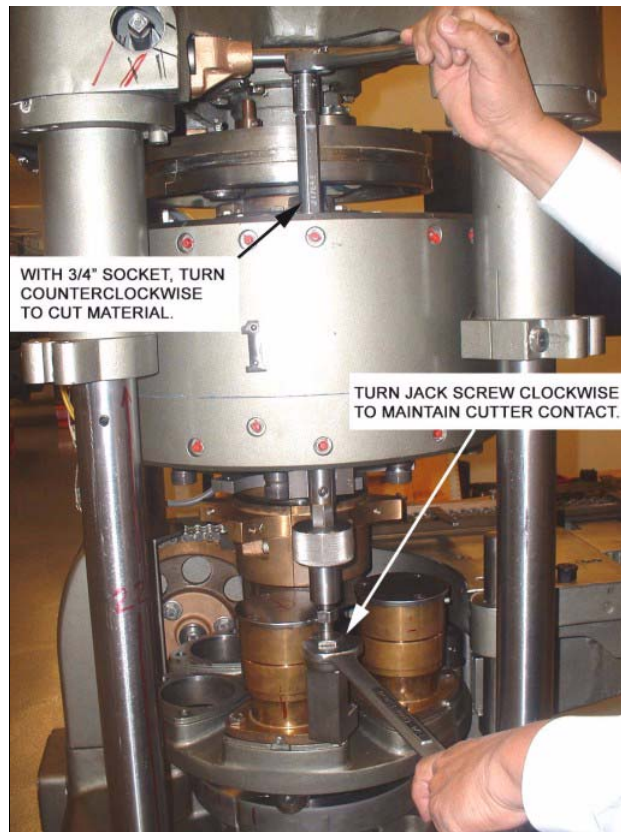
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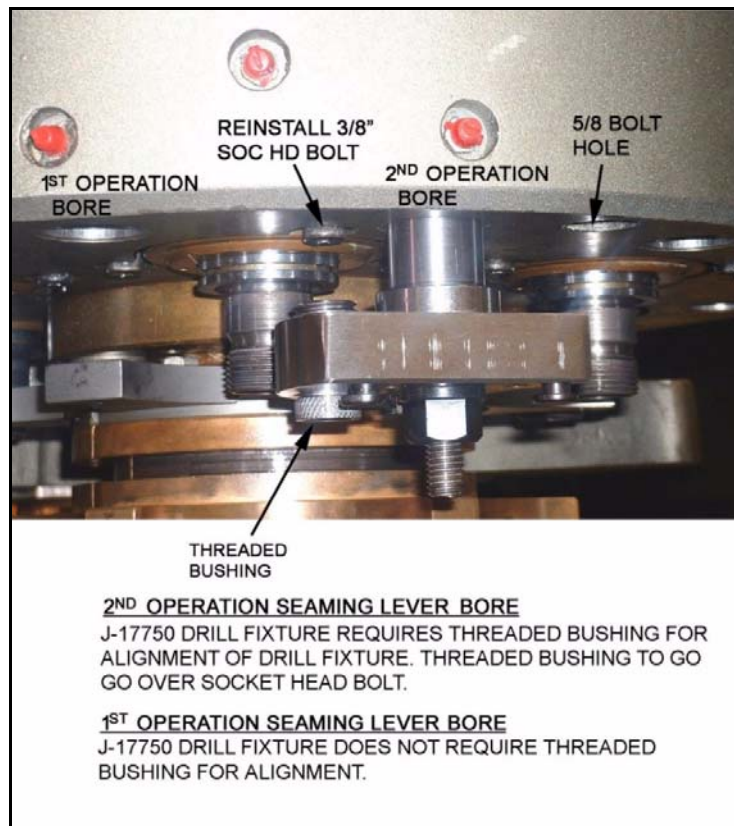
(3)



(4)



(5)



Subject:

60L Seamer Modification Required On 2L620 Seaming Cam Section (for size 200MS & up).

Purpose:

To modify the 2L620 seaming cam in order to provide additional clearance to the 2nd operation seaming rolls in the off seam position.

Benefit:

The seaming cam modification gives a true reading of the 1st operation seam thickness on the "off seam position", otherwise without the modification the 2nd operation seaming roll will contact the 1st operation seam.

Description:

The existing 2L620 seaming cam can be modified into 75L620 seaming cam (with offset) by replacing;

- 129L616 bushing with 386L616 bushing
- 30L628 seam cam eccentric pin with 66L628 seam cam eccentric pin.

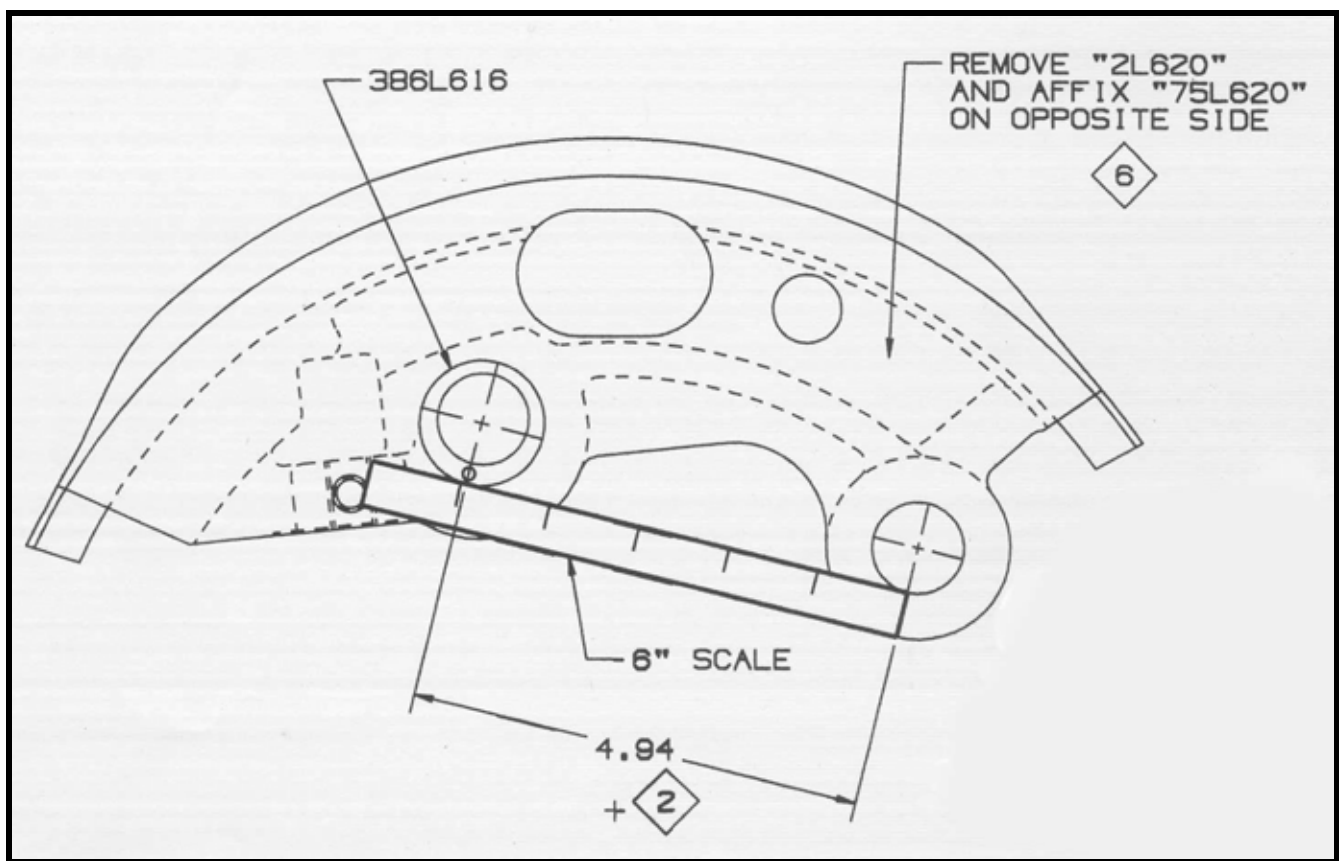
The 75L620 seaming cam is the same as 2L620 seaming cam with the exception of the eccentric bore with an offset incorporated within.

Parts Needed:

To Modify 2L620 Seam Cam	75L620 Seam Cam (with offset)
(1) 386L616 Bushing (with offset)	(1) 129L616 Bushing
(1) 66L628 Eccentric Pin	(1) 66L628 Eccentric Pin

Installation Procedure:

- 1) Remove 129L616 bushing from existing 2L620 cam.
- 2) Use a scale as shown and mark the 4.94" dimension on the face of casting.
- 3) Locate and align the "O" marked face of 386L616 bushing on the marked face of the casting.
- 4) Press the bushing flush onto seaming cam.
- 5) Install the seaming cam on the seamer and use 66L628 pin.
- 6) When replacing 2L620 cam for this application, order 75L620 cam, 129L616 bushing, and 66L628 pin.



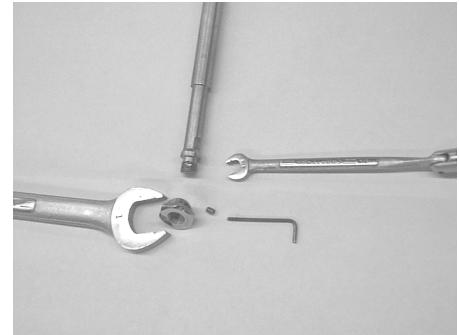
Ref. Engr. Bulletin #75 & Data Sheet G60.243

Model 60L Knockout Pad Removal Procedure

(For threaded Knockout Pads and Rods only)

Tools Required;

- 3/8 open end wrench
- 1" open end wrench
- 3/32 Allen wrench



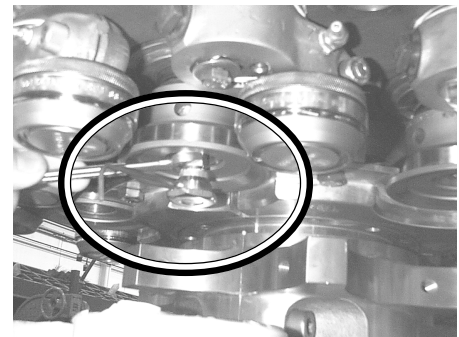
Tools required

1. Align Seaming Station beneath the Seaming Cam opening.



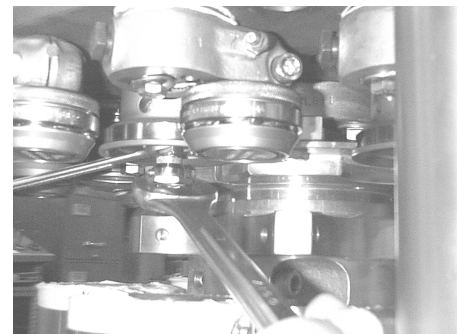
Seaming Cam Opening

2. Push down on the Knockout Rod through the opening and insert the 3/8" open end wrench onto the bottom of the Knockout Rod above the Pad.
3. Locate the #10 set screw on the side of the Pad and loosen using a 3/32 Allen wrench.



4. Use a 1" open end wrench to remove the Pad.

CAUTION: Pad and Rod threads are left hand. When replacing the Pad, tighten only enough to align the set screw with the drilled hole in the Rod for locating the Screw.



Model 60L Seaming Spindles

Scope:

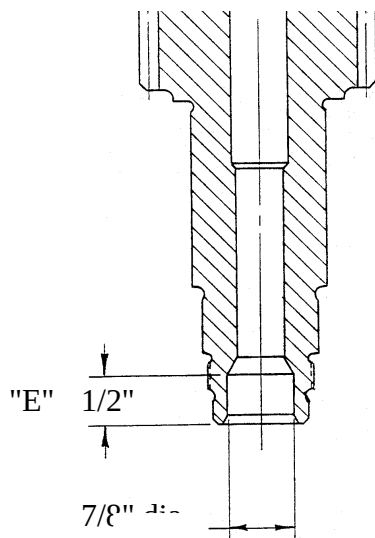
Seaming Spindle modifications made on Data Sheet G60.193 for Splined Knockout Pads

Description:

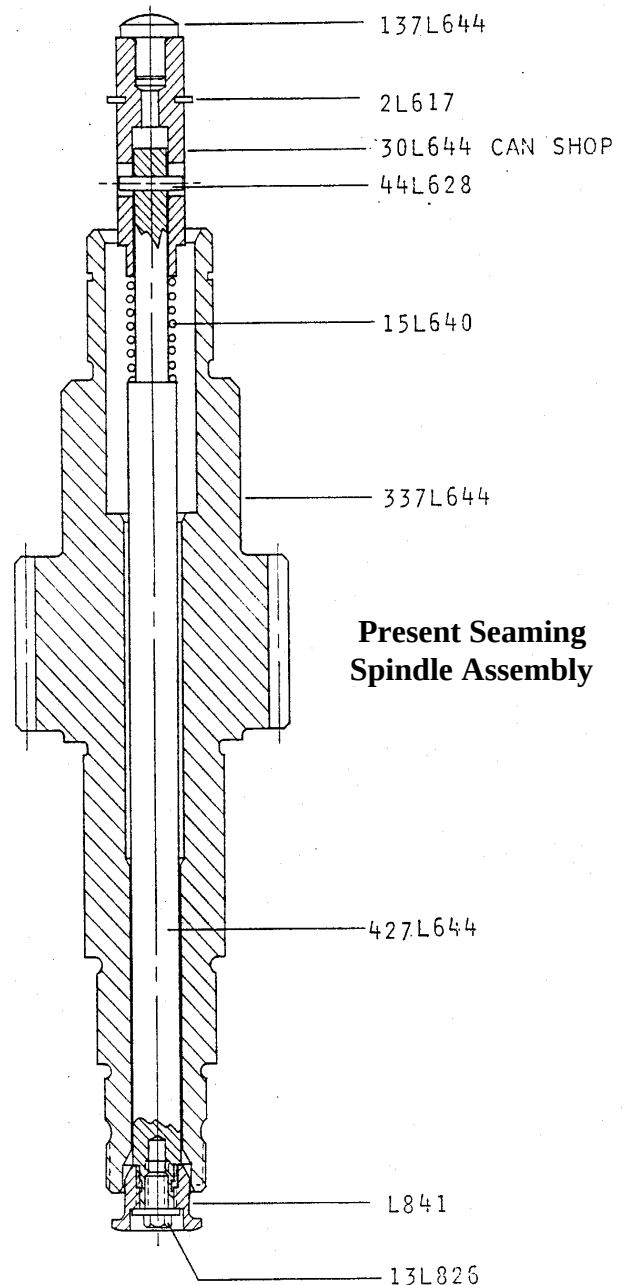
In September 1986, the 337L644 Seaming Spindle was changed to cut the inside bore on the threaded end from 11/32 to 1/2 inch deep. This change was made to provide clearance for the L841 Splined Knockout Pad with the 427L644 Knockout Rod.

Seaming Spindles manufactured prior to this will not have a deep enough bore to accept the larger Splined Knockout Pads.

See sketches below.



"E" dimension changed 9/86
from 11/32" to 1/2" deep.



**Present Seaming
Spindle Assembly**

Subject:

60L Seamer Two Piece Seaming Spindle Seal Sleeve

Purpose:

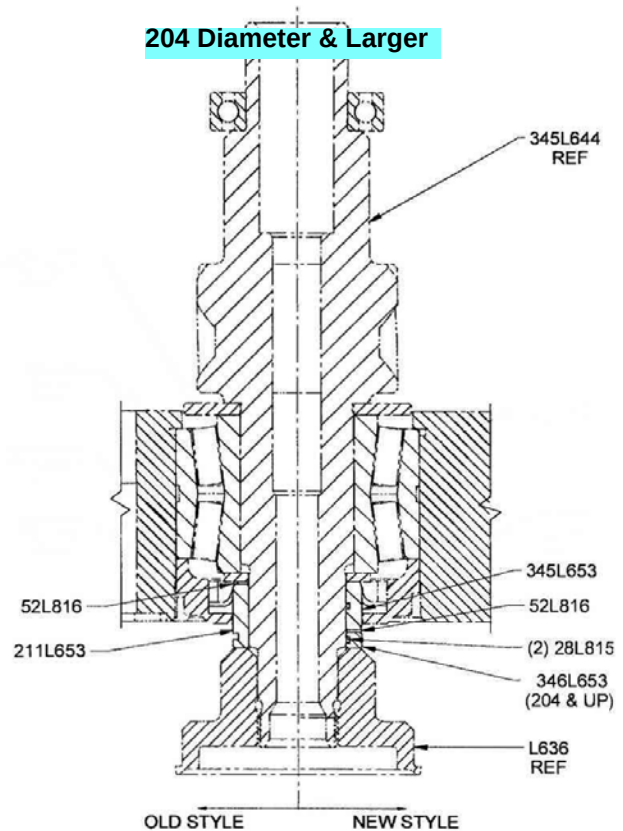
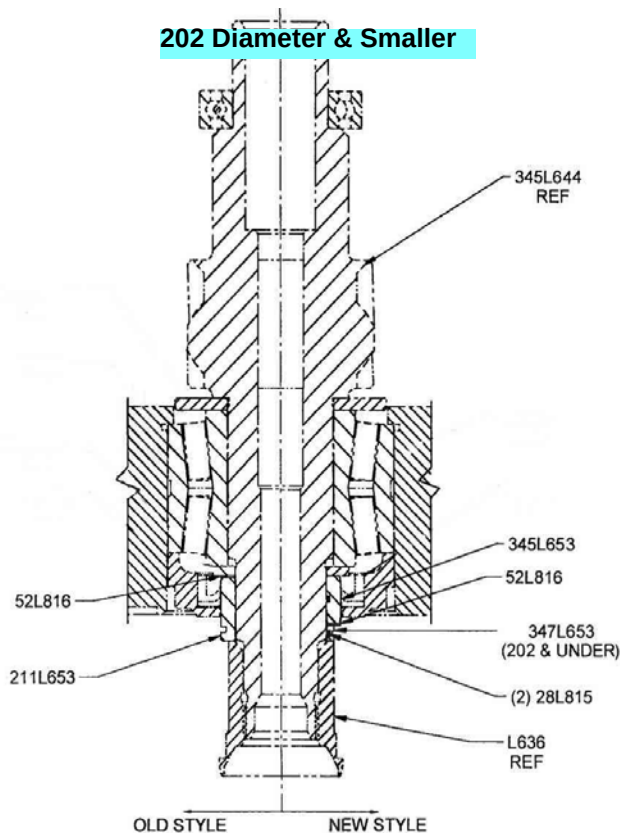
A change in design to a two-piece seal sleeve will reduce time when shimming seaming chucks to desired dimension.

Benefit:

- Eliminates the removal of the seaming chuck bell when shimming
- Reduces time required to shim the seaming chucks
- Increases the life of the seal, as the seal sleeve top will remain in place during shim adjustments

Description:

202 Diameter & Smaller		204 Diameter & Larger	
Old Design	New Design	Old Design	New Design
211L653 Seal Sleeve	345L653 Seal Sleeve Top	211L653 Seal Sleeve	345L653 Seal Sleeve Top
	347L653 Seal Sleeve Bot		346L653 Seal Sleeve Bot
	(2) 28L815 O-Rings		(2) 28L815 O-Rings



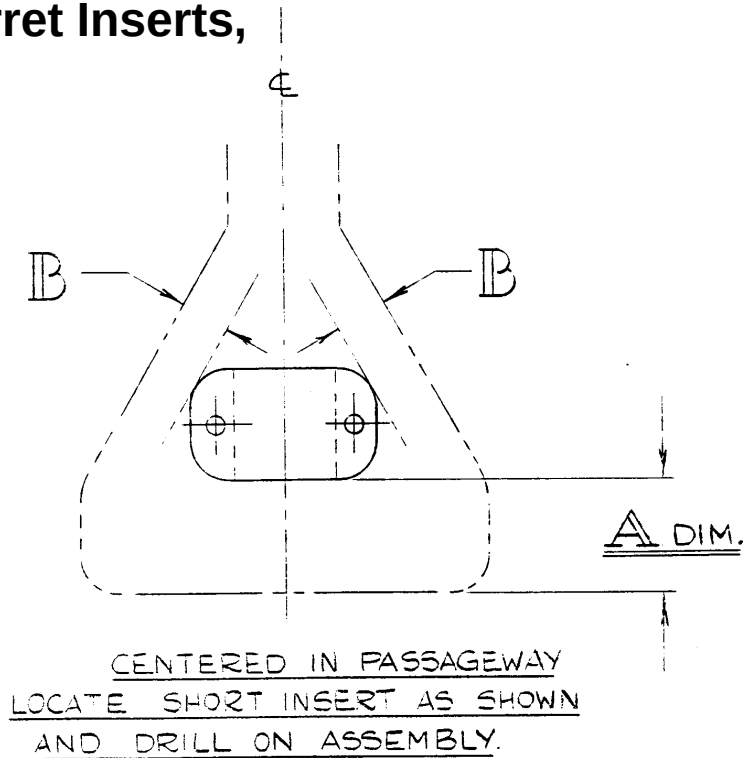
Ref. Engr. Bulletin #100



Model 60L Cap Feed Turret Inserts, Short 184L670

Insert Installation Procedure

Can Diameter	'A' Dim.	'B' Dim.
404		5/16"
401		5/16"
307		5/16"
303		5/16"
300		5/16"
211	3/4"	



Drill Inserts using a #32 drill (.116) x 1/2" deep.
Follow this drill with a 1/8" drill bit.
DO NOT DRILL THROUGH TURRET.
Install 1/8" x 7/16" stainless steel dowel pins.

Use 5/16" diameter dowel pins
to locate the Inserts on all can
diameters (see photo) except the
211 diameter or smaller.

